Lecture 19: Overfitting I

- Remember the Ptolemaic (geocentric) model that was able to explain Mars' retrograde motion (illusion that Mars changes the direction of its trajectory in the sky) just as well as the Copernican (heliocentric) model in the night sky.



- (see `lecture9-mars-geocentric.pm4` file)
- But even Copernicus model wasn't entirely true, since he was committed to circles. Actually very complicated and very wrong...
- (see `lecture19-copernicus-circles.mp4` file)
- Actually very complicated and very wrong...still a Fourier approximation of the solar system.

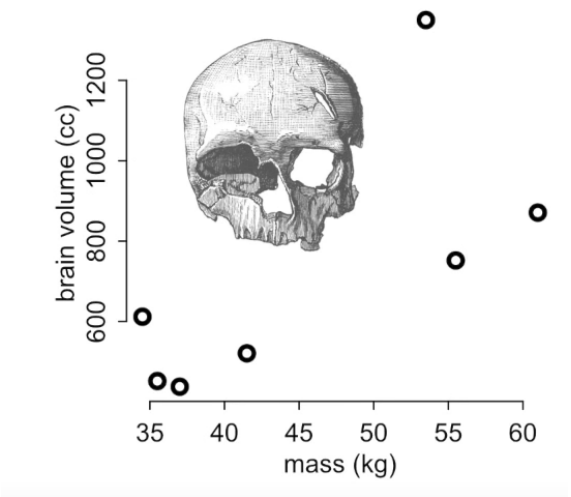
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Causal modelling versus approximate (but accurate) predictions

- What is ingenious in the Copernican model is that it renders many observations: the Sun is not exactly at the centre of Earth movement, planets accelerates when they are closer to the Sun, etc.
- In scientific modelling, we are always dealing with
  - causal inference: what is the structure of the system, what are the causes of the phenomena we observe
  - for the other part of the system that do not have good causal structure information, how do we accurately approximate the causal structure so that we can move forward, e.g. making predictions.
- Good news: we can move forward.
- Bad news: the tools that we used for causal modelling are not always those we encounter to do good causal approximations.

Problems of predictions

How to construct good approximations to the unknown causal structure?



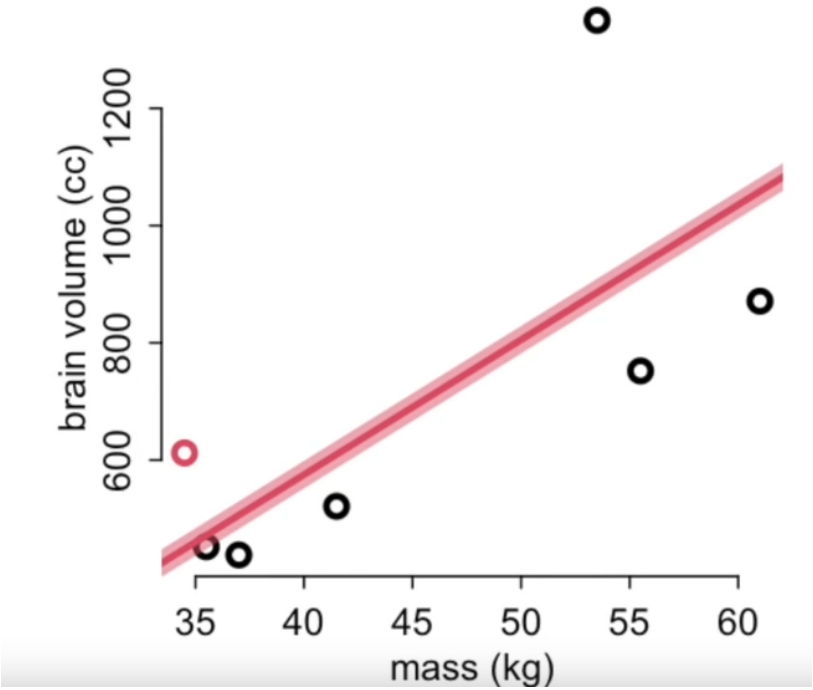
- What function describes these (7) data points from fossil hominin? ("compression" algorithm in fitting a "simple" model)
- It is different to asking what explains the points (causal inference; evolutionary history of these species).
- We might also ask what is going to happen if we changed a point's mass (counterfactual intervention).
- Finally, today: what is the next observation from the same process. This is (one meaning of the word) *prediction*.

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Good predictions do not require good causal inference

Leave-one-out Cross-Validation (LOO-CV)

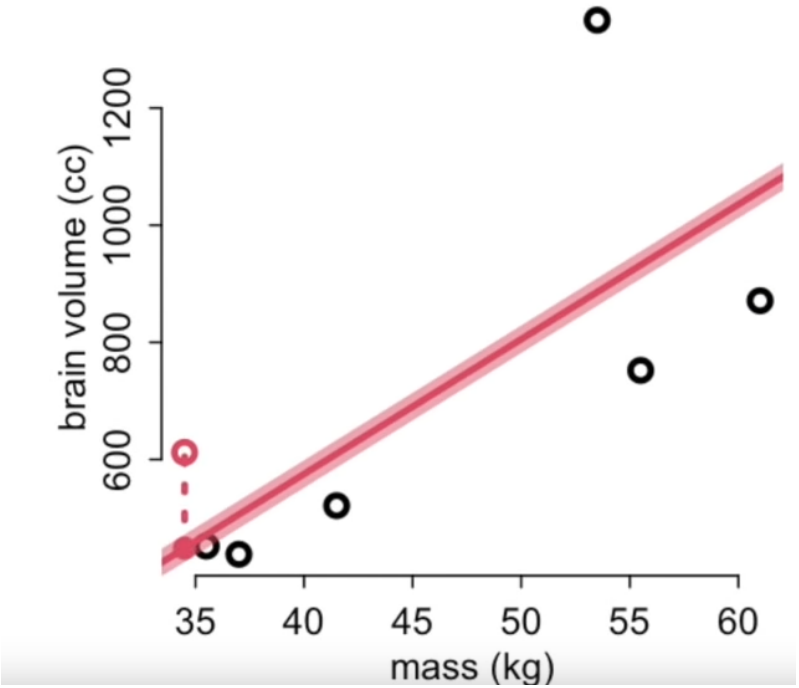
1. Drop one point.
2. Fit line to the remaining points.
3. Predict the dropped point.
4. Score is error on dropped.
5. Repeat (1-4) with the next point and average.



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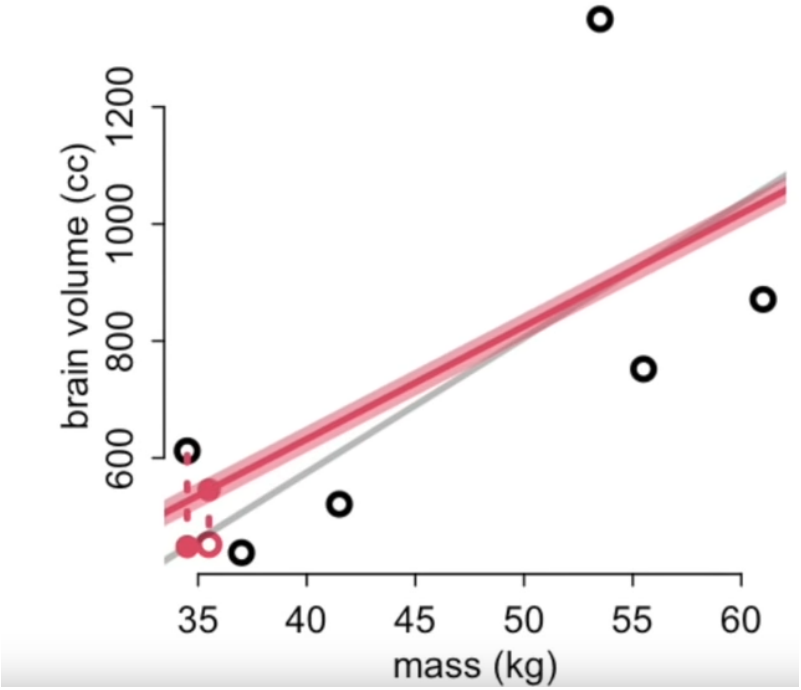
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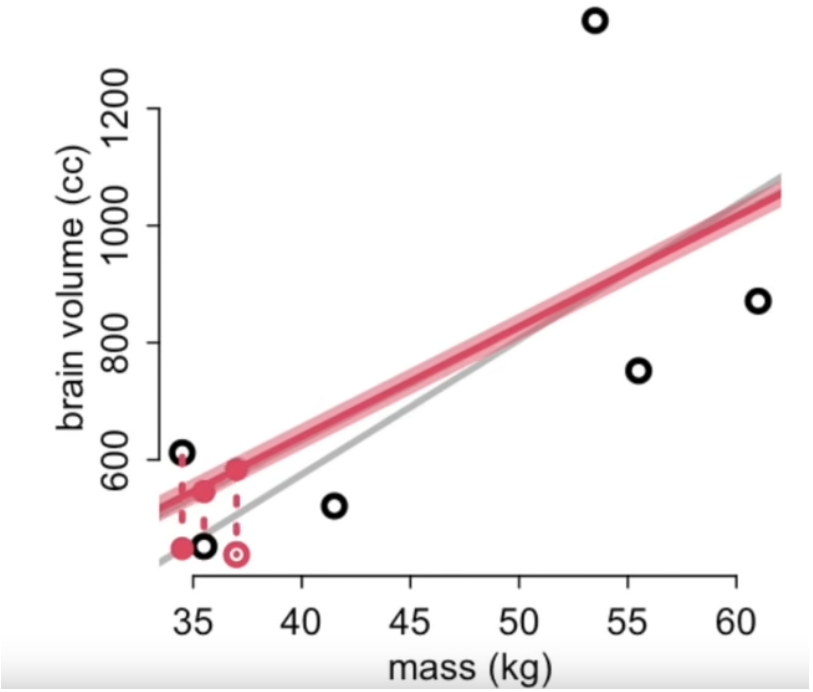
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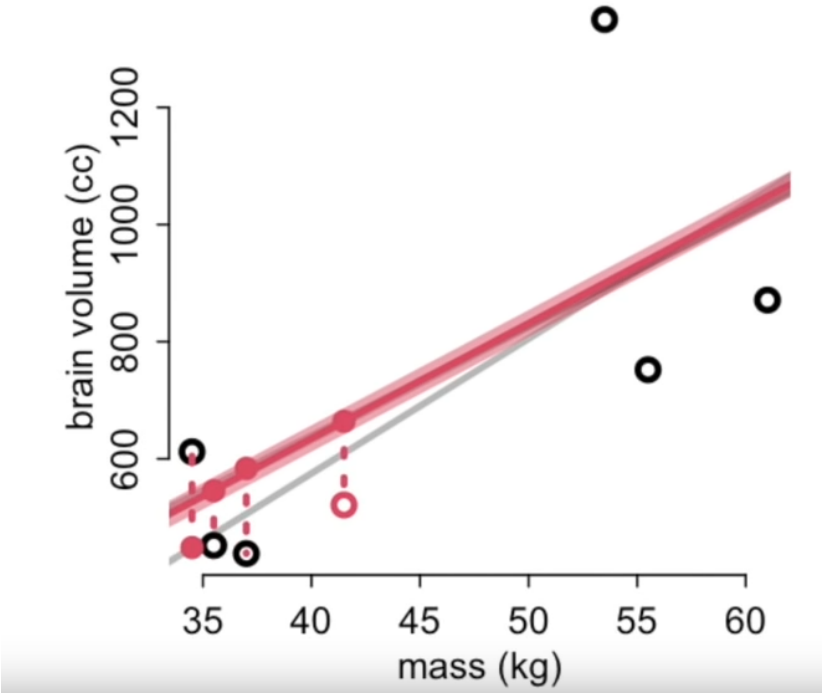
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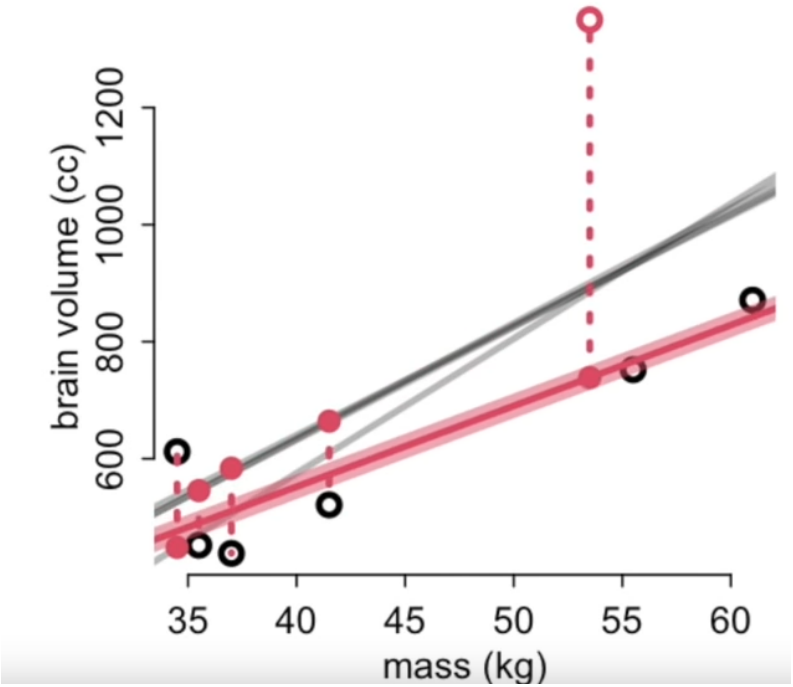
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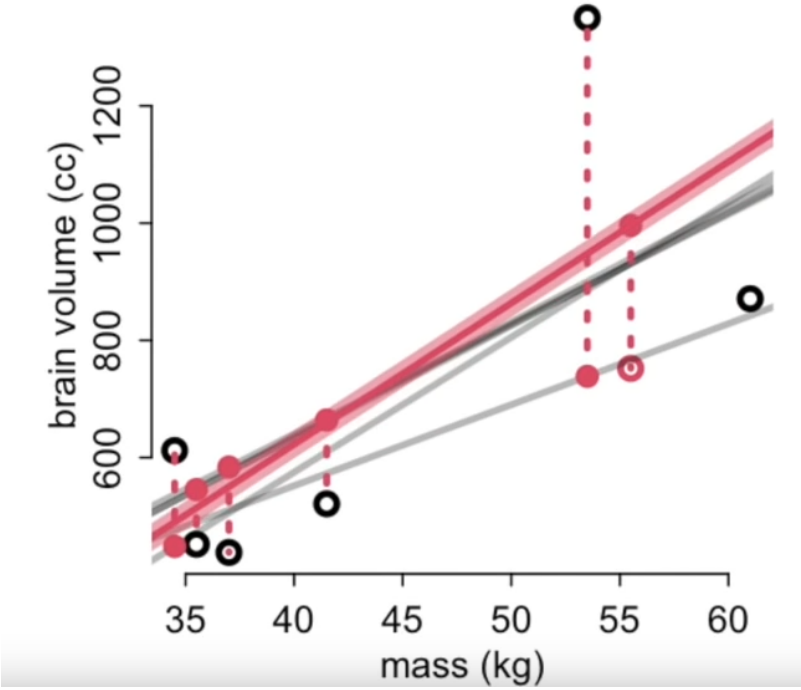
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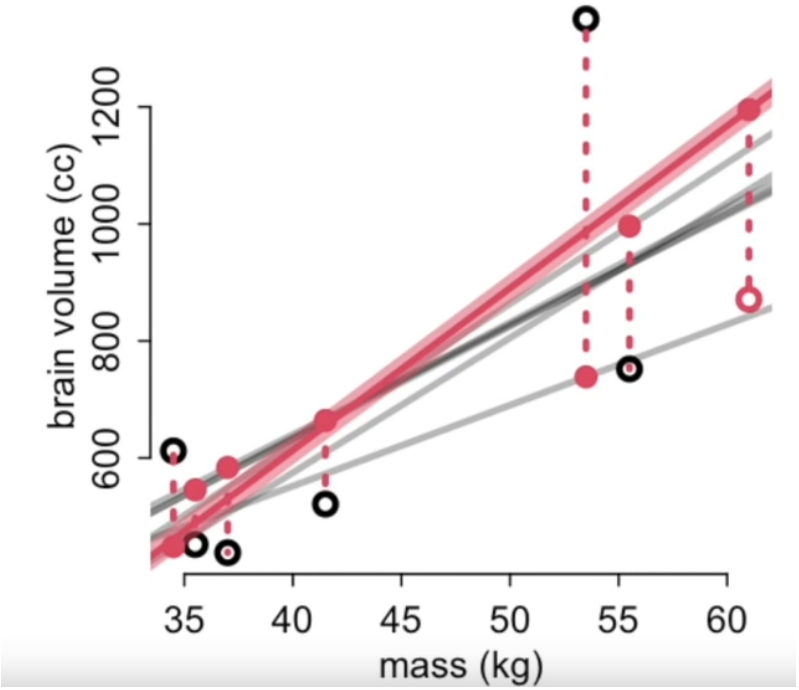
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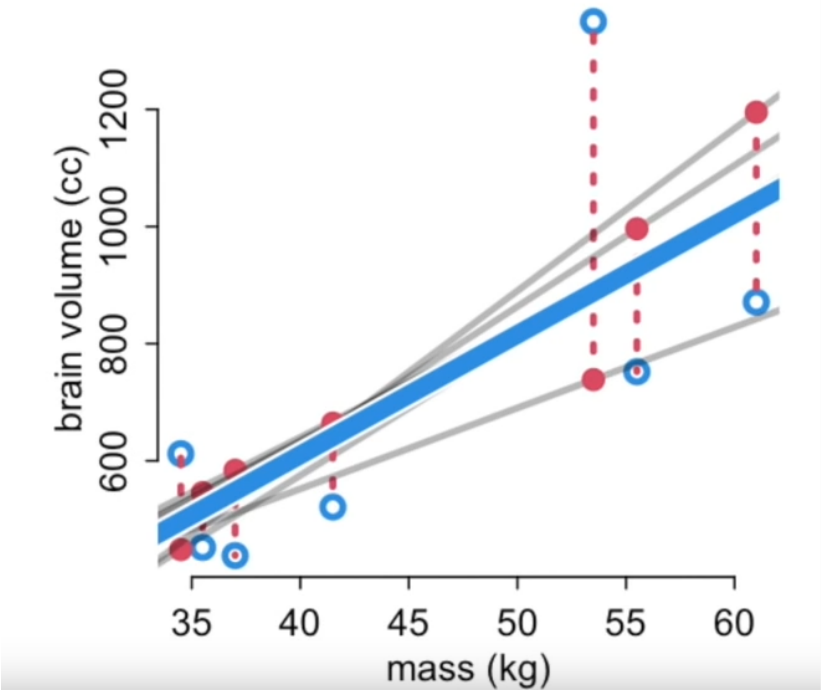


Good predictions do not require good causal inference

Leave-one-out Cross-Validation (LOO-CV)

1. Drop one point.
2. Fit line to the remaining points.
3. Predict the dropped point.
4. Score (Sum squares) is error on dropped.
5. Repeat (1-4) with the next point and average.

- Don’t score with the "blue errors".
- You have "In" score (points used to build the line) and "Out" score (dropped point).
- In general "Out" score > "In" score.



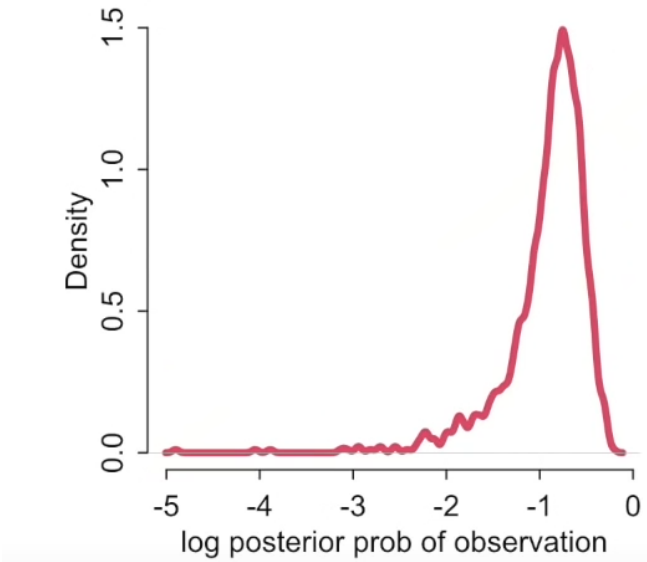
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Bayesian Cross-validation scoring

- We use the entire posterior, not just a point prediction.
- Cross-validation score (log point-wise predictive density):

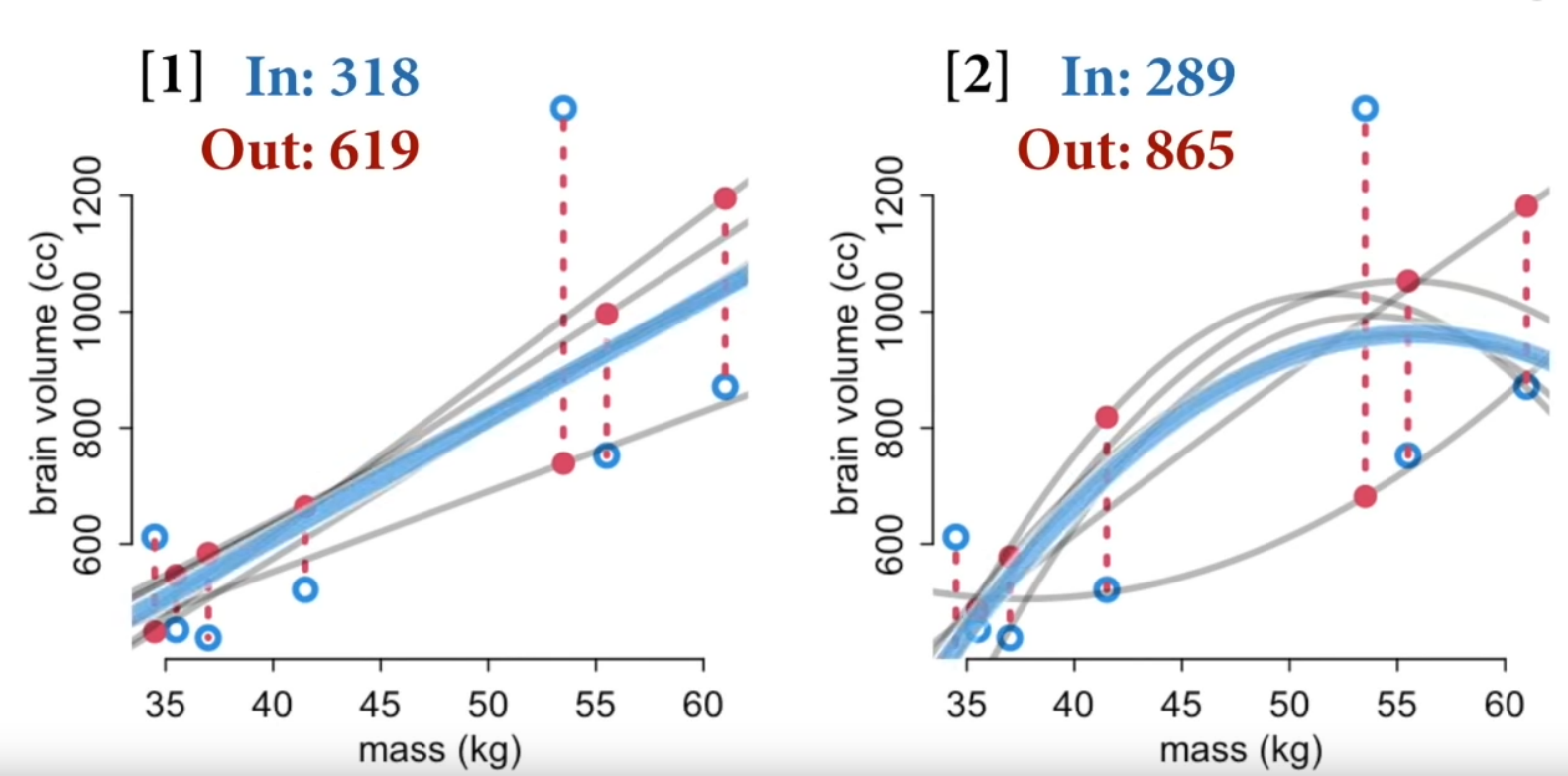
$$\text{lppd}_{\text{CV}} = \sum_{i=1}^{N} \frac{1}{S} \sum_{s=1}^{S} \log \left[P(y_i | \theta_{-i,s}) \right]$$

- N data points, S samples from the posterior.
  $\log \left[P(y_i | \theta_{-i,s}) \right]$: log-probability of each point i computed with the posterior that omitted that point.
  Then averaged over the S posterior samples (integral approximation)
- (see details in the book)



Back the skull vs mass modelling

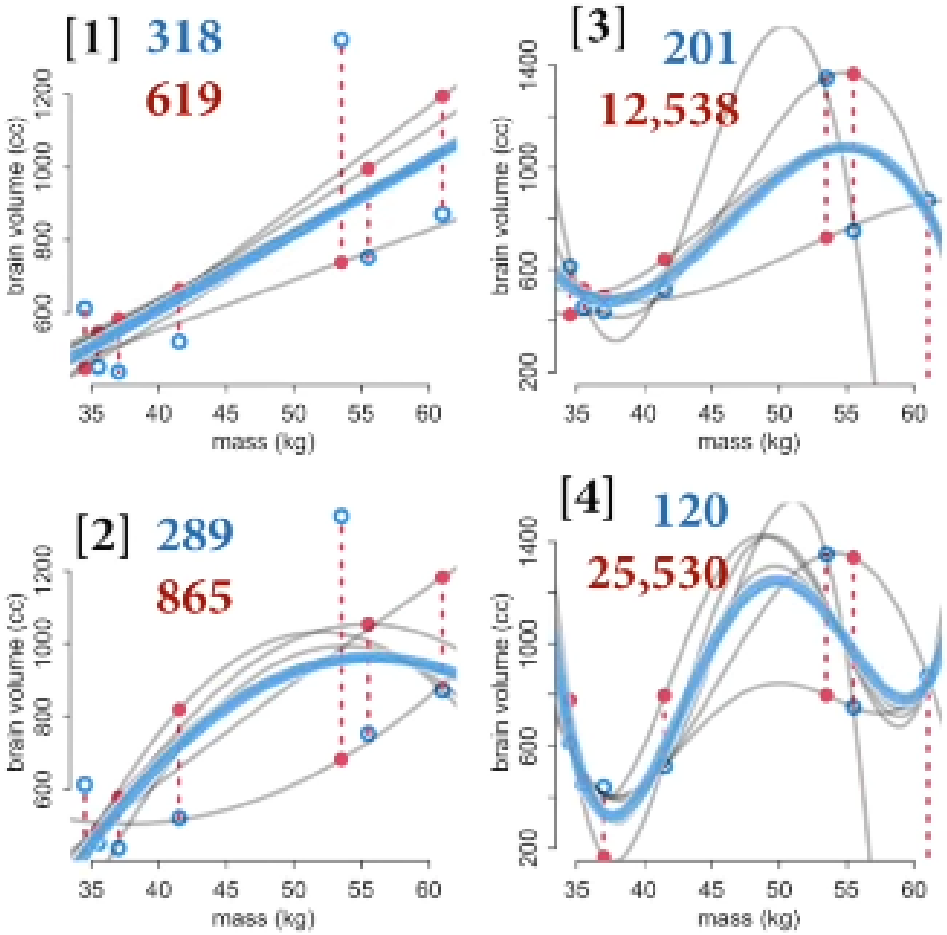
Another model

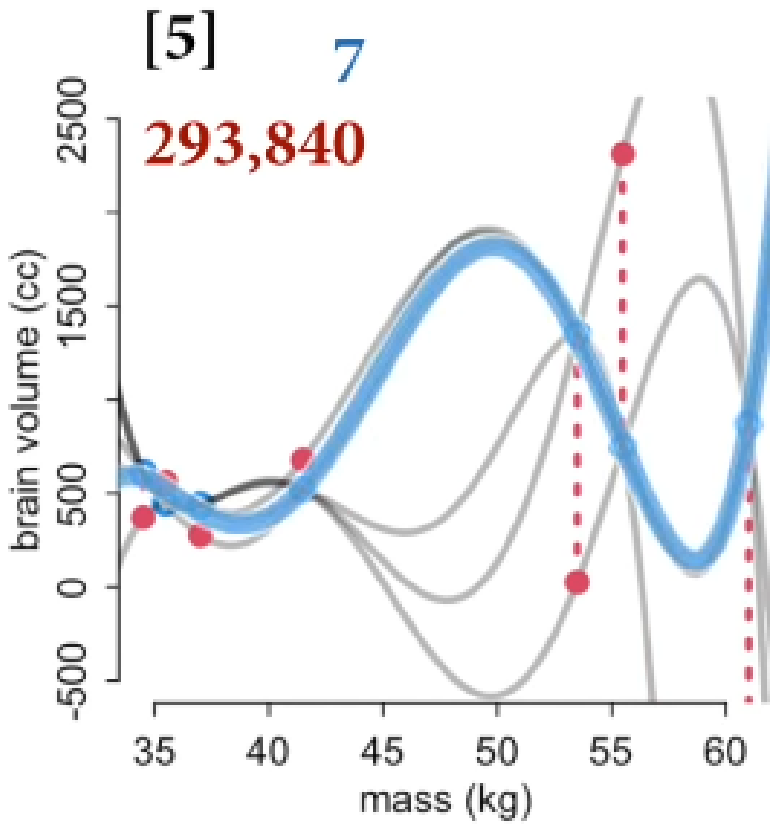


The more complex model has a lower In-score (error), but its Out-score has gotten worse...

Back the skull vs mass modelling

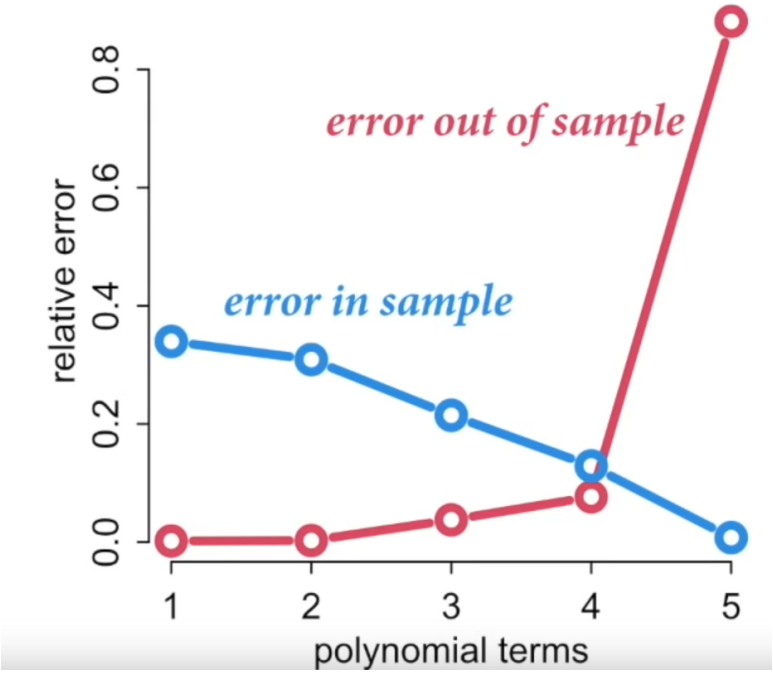
More complex models are...stupid





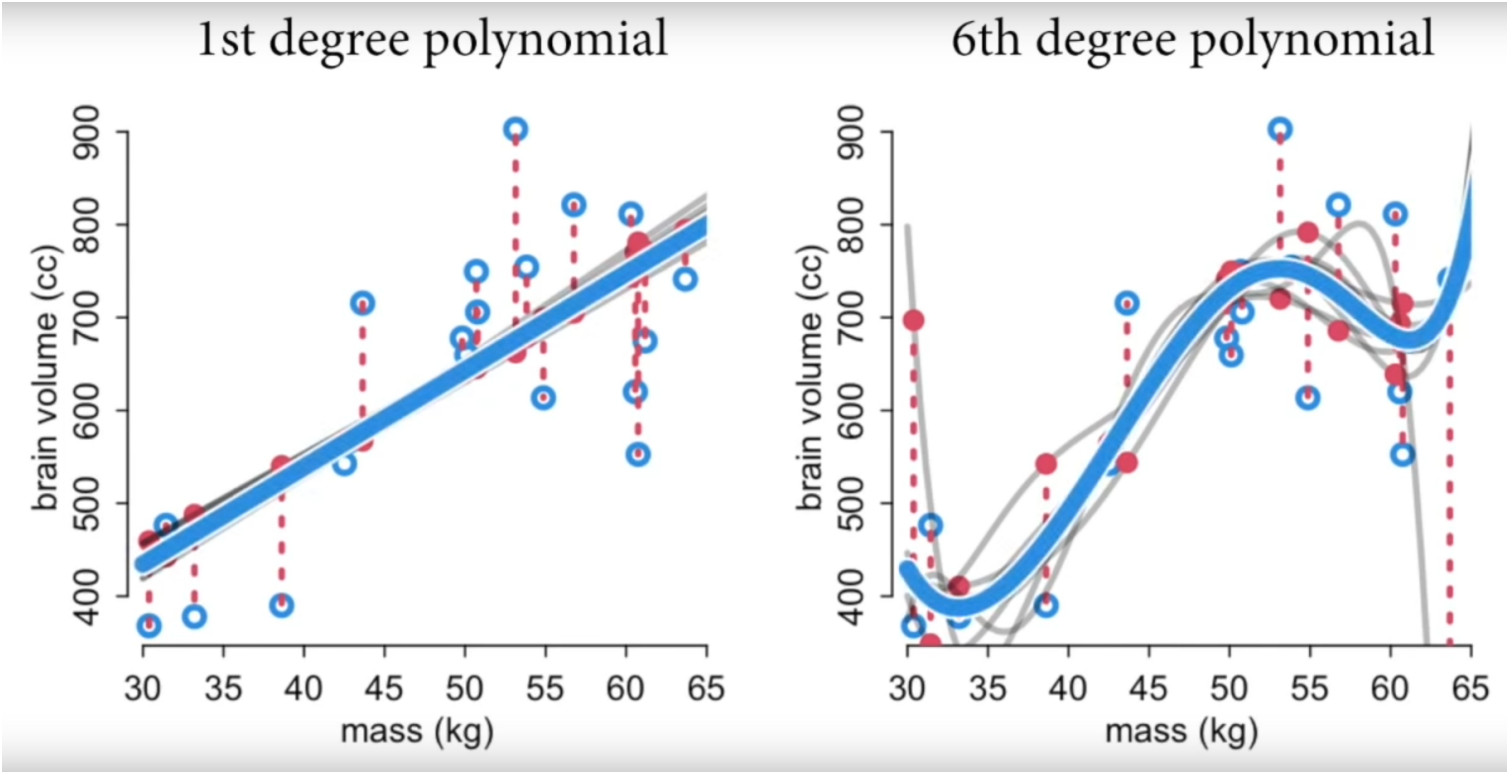
Cross-validation

- For simpler models, increasing the number of parameters improves the fit to the sample.
- But it seems to reduce the accuracy of the out-of-sample predictions.
- Most accurate models trade off flexibility (complexity) and overfitting.



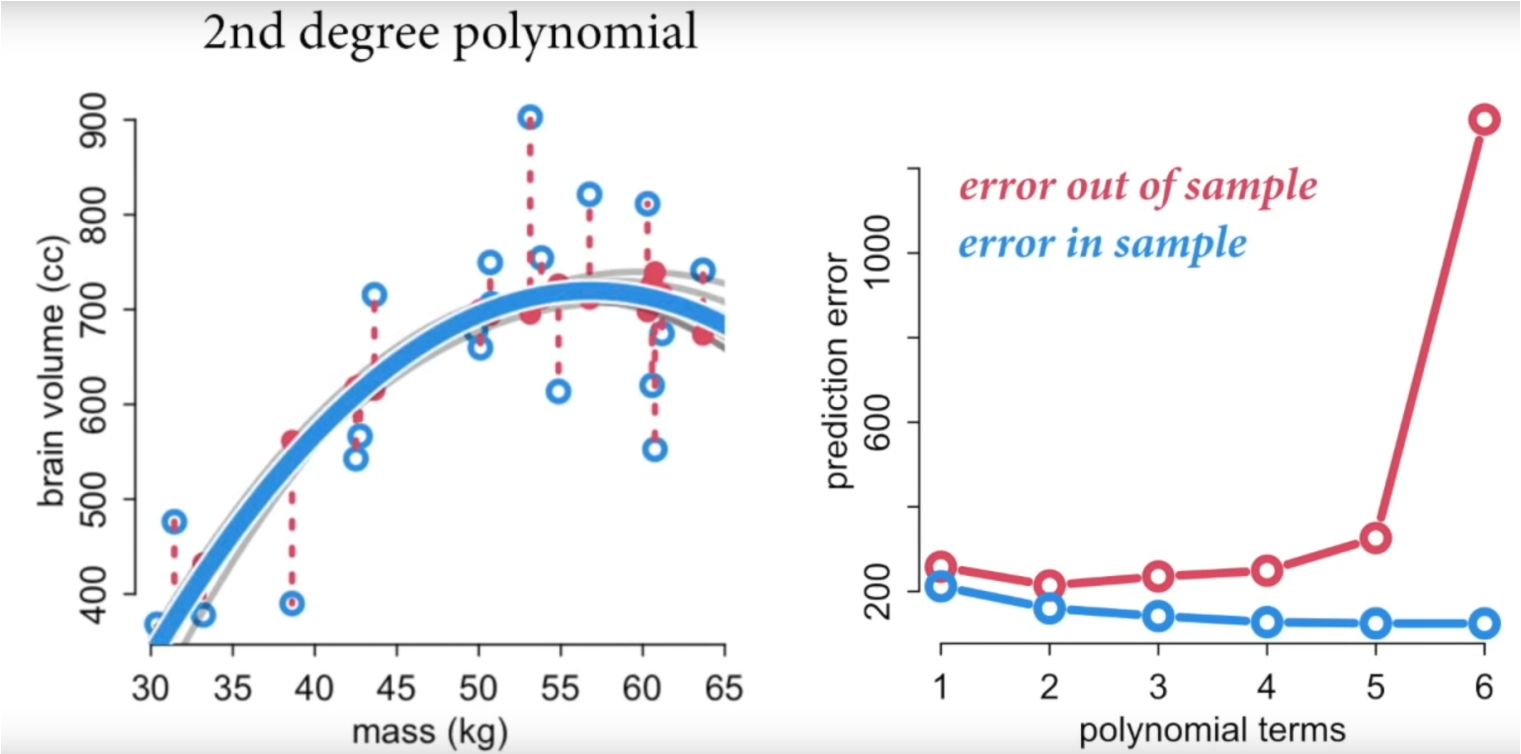
(relative error is error divided by the sum)

An example with mode data points



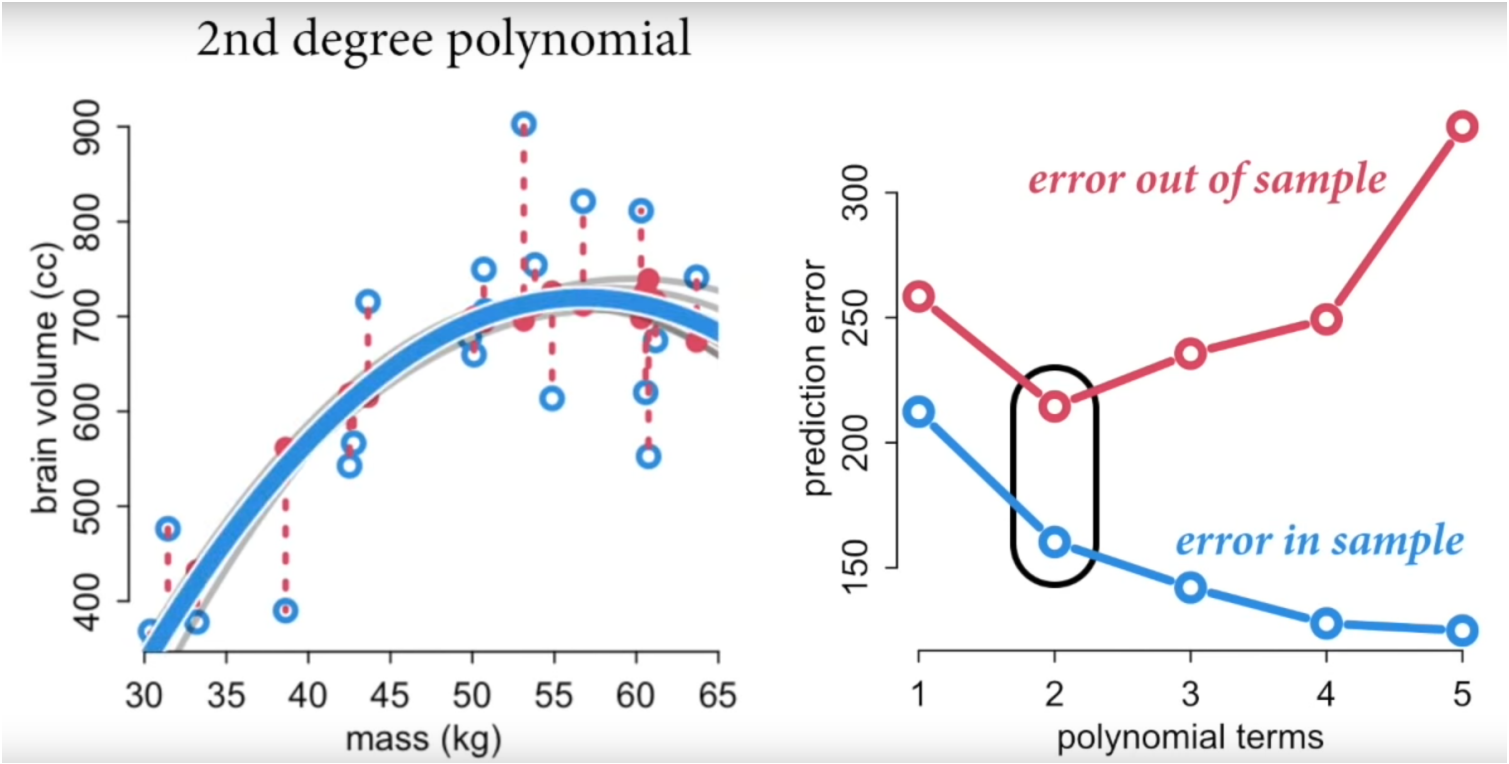
Notice the many similar models on the left and the 6th-order polynomials are quite different and fit the in-sample quite well.

An example with mode data points



Now the 2nd order polynomial is best with that many points. It does not mean it is causally more relevant.

An example with mode data points



Now the 2nd order polynomial is best with that many points. It does not mean it is causally more relevant.
Another slogan: fitting is easy, predicting is hard.

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Regularisation

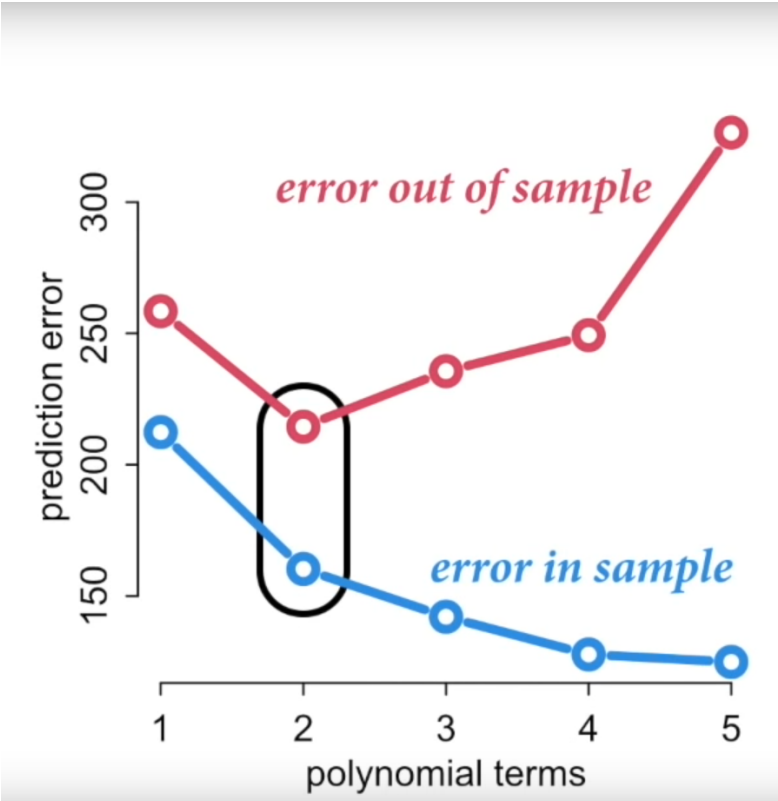
Cross-validation was measuring overfitting. It doesn't *do* anything about it.

- When designing a model, we notice(d) that overfitting depends on the priors.
- "Sceptical" priors have tighter variance and reduce flexibility.
- *Regularisation*: seeks regular features of the process.
- Good priors are often tighter than you think.
- A good model will discard irregular features.



Just like Olaf, we should be sceptical until enough evidence arrives to justify the complexity of the patterns.

Regularisation, effect of the priors



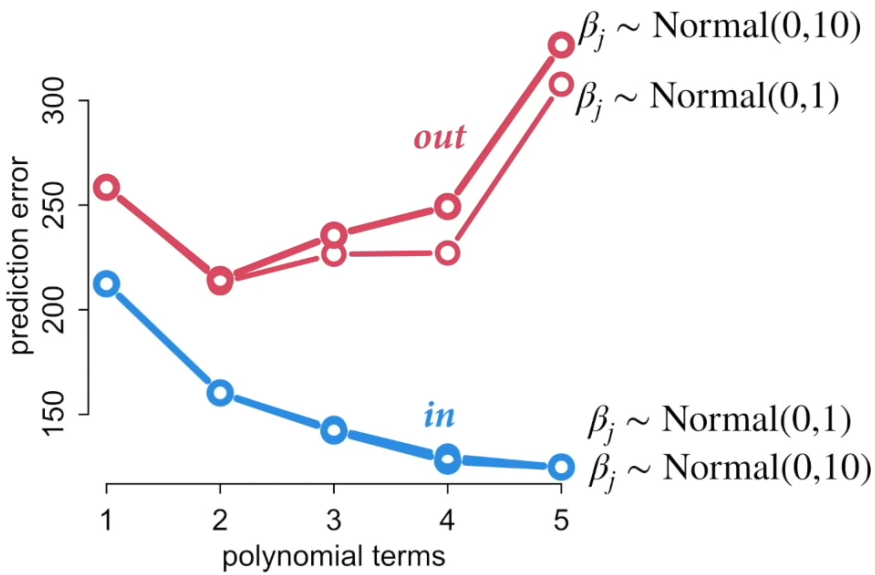
$$\mu_i = \alpha + \sum_{j=1}^{m} \beta_j x_i^j$$

$$\beta_j \sim \text{Normal}(0, 10)$$

It's a very wide prior, essentially flat from the point of view of standardised variables.

Regularisation, effect of the priors

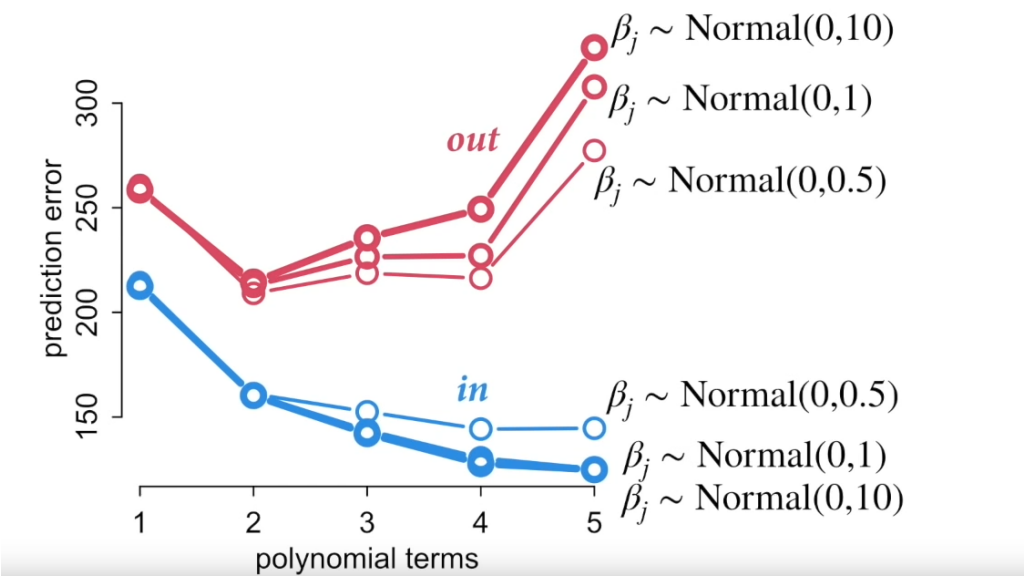
More sceptical



Better, we have reduced the "out" prediction error by reducing the flexibility. On all models, not so much on the (still) best model, the polynomial of order 2.

Regularisation, effect of the priors

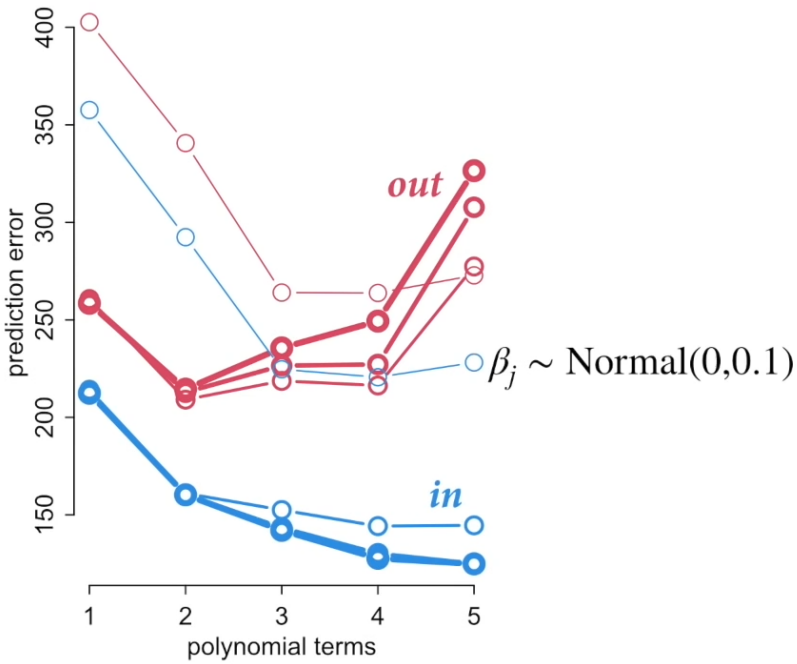
Very sceptical



We can make it even tighter. Another improvement. The "In"-sample fit is now worse. And the "Out"-sample better. And now the 4th order polynomial is the best model from the prediction perspective.

Regularisation, effect of the priors

Too sceptical



Now we are over-regularising...only the 5th order polynomial performs better than its version with looser priors. So the amount of needed regularisation seems to depend on model complexity.

A good news is: it is almost always possible to do better than a flat prior.

Conclusive remarks

on prior regularisation

- How to choose the width of a prior?
  - For causal inference, use science or common sense (we have already done that).
  - For pure predictions, you can tune the parameter using CV.
- In practice, many tasks are a mixture of inference and predictions. Ideally, we want to know about the system and predict its behaviour. Example of the geocentric vs heliocentric models.
- No need to be perfect, doing better than flat priors is already a gain. There is no perfect model either, btw.